

ERSAL SCIENCE & COMMERCE ACADEMY

Bar Lab Nehar Sitara Colony # 2, Chungi Amer Sidhu Lahore. 0322-4540241

Father's Name _____

Roll #: _____

Class _____

Section _____

Subject INORGANIC

Date _____

	MCQ's	Short Questions	Long Questions					Grand Total
			Q1	Q2	Q3	Q4	Q5	
Total Marks								
Obtained Marks								

(A) (B) (C) (D)

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Short (2015)

(1)

covalent bond
double or triple

①

Bent Bond.

— Bent bond is also known as

"Banana bond" is a term in chemistry

that refers to a type of covalent

chemical bond with geometry somewhat

reminiscent of a banana.

The term itself is the general

representation of electron density or

organization of a representation

of double or triple bonds within

a compound that is an alternative to

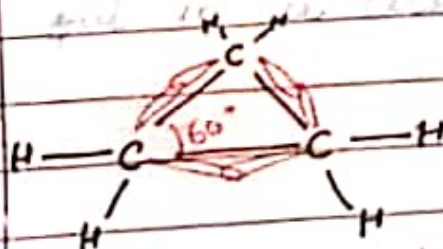
sigma and pi bond model.

Bent bond is formed by an oblique

overlap of atomic orbitals as a result

of which the resultant bond angle

and is in molecular fragments.



Banana bond in cyclopropane



Applications of Lanthanides:

Following are the major applications of Lanthanides:

- The best use is for the production of alloy steel for plates and pipes.
- A well known alloy is misch metal ^{which consists of lanthanide metal (95%) iron (5%) and traces of S, Ca, C & Al.} which consists of lanthanide metal (95%) iron (5%) and traces of S, Ca, C & Al. A good deal of misch metal is used in Mg-based alloy to produce bullet shell and lighter flint.
- Mixed oxides of Lanthanides are employed as catalyst in petroleum refining.

- Lanthanides are used in lamps, laser magnets, phosphors, motion picture projection.
- Some alloys containing neodymium and cerium make excellent magnets.

Limitations of VB Theory:

- This theory is unable to explain the paramagnetic behaviour of Δ -molecules.
- This is unable to explain bonding in electron deficient molecules.
- It can't explain the non-existence of noble gas molecules.
- It can't describe the proper shape and geometry of molecules. (e.g. unable to differentiate b/w the square planar & tetrahedral geometry)

... rule.
... for the formation of
... the isolated gaseous
... transition metals.

It provides a quantitative justification for the sequence of orbitals in energy level diagram.

3d helps to explain the filling of ns-orbital (4s, 5s, 6s etc --- orbitals) prior to the filling of (n-1)d orbitals (3d, 4d, 5d, etc).

Shost (2016)

(1)

Limitations of VESPER Theory:

Although this theory is very useful in explaining geometries of a large no. of molecules or ions, yet it has following limitations

- This theory can't explain the shape of molecule having any polar bonds
eg H_2O should have the same structure as CO_2 but in fact it is linear.
• It is unable to explain the shapes of molecules having delocalized π -electron systems.

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Section

Subject**Date**

B	C	D
E	C	D
F	C	D
G	C	D
H	C	D
I	C	D
J	C	D
K	C	D
L	C	D
M	C	D
N	C	D
O	C	D
P	C	D
Q	C	D
R	C	D
S	C	D
T	C	D
U	C	D
V	C	D
W	C	D
X	C	D
Y	C	D
Z	C	D

Q. Explain the shape of certain molecules which have inert air of reactions.

(2)

Comparison blue

Armi orbital

Molecular Orbital

Atomic orbital is the region having the highest probability of finding an electron in an atom.

→ Molecular weight is an
important factor in
determining the
binding of a molecule.

→ Formed by the electron
lost against the atom

→ Point of the previous
atoms, carbons that
have nearly same energy

→ Shape is determined by the type of atomic orbital (s, p, d, etc.)

→ Shape is determined by the shape of atoms & bonds that make the molecule.

→ Schrödinger equation

3. 10. 1

Fajan's rule:

Fajan's rule formulated by Kazimierz Fajan's in 1923, are used to predict whether a chemical bond will be covalent or ionic, and depend on the charge on the cation and the relative sizes of the cation and anion.

OR

— Polarization power of a cation is usually called its "charge density" or "ionic potential". It is represented by ϕ and mathematically expressed as

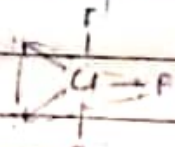
$$\phi = \frac{\text{charge of cation}}{\text{radius of cation}}$$

(4)

Some molecules form T-shapes instead of trigonal planar shape:

The species having three bonding e⁻ pairs and two lone pairs of e⁻ (bp = 3, lp = 2) have 'T-shaped bent geometry' instead of trigonal planar because both the lone pairs occupy the equatorial positions of the trigonal bipyramidal geometry. This position of lone pair involves minimum repulsions as compared to axial positions.

e.g. ClF3, BH3, ICl3 etc.



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2 (B) (C) (D)
3 (A) (B) (C) (D)
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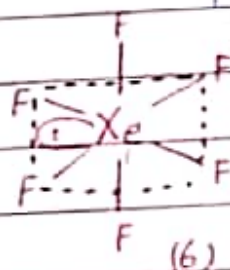
(5)
Hybridization in XeF_6 molecule.

$\text{Xe} = 54$

Xe (ground state) = $[\text{Kr}] 4d^10 5s^2 5p^6$

Xe (excited state) = $[\text{Kr}] 4d^10 5s^1 5p^5$

Xe (hybridized) = $[\text{Kr}] 4d^10 5s^1 5p^5$
 $3p^1 d^2$ hybrid



Irregular octahedral geometry.

Effective nuclear charge experienced by a valence p -e in boron.

$$sB = 1s^2 \cdot 2s^2 \cdot 2p^1$$

$$Z = 5$$

$$\sigma = 0.35 \times 0 + 0.85 \times 4 = 3.4$$

$$Z_{\text{eff}} = Z - \sigma = 5 - 3.4 = 1.6$$

$$Z_{\text{eff}} = 1.6$$

(1)

Uses of lanthanides:

Repeat

(2)

Application of inorganic compounds in biological systems:

Complexes like hemoglobin, chlorophyll, vitamin B₁₂, etc. are naturally occurring.

Chlorophyll, an important constituent of plants, is a chelate compound containing Mg^{+2} in the centre linked with pyrrole nuclei.

Hemoglobin, the red colouring pigment of blood, is an iron chelate responsible for oxygen transport in the living systems.

(3)

Calculate Z_{eff} for 4s e⁻s in Cobalt ($Z=27$)

$$_{27}Co = \underbrace{1s^2, 2s^2, 2p^6}_{10} \underbrace{3s^2, 3p^4, 3d^7}_{15} \underbrace{4s^2}_2$$

$$\sigma = 0.35 \times 1 + 0.85 \times 15 + 1.0 \times 10$$

$$\sigma = 23.10$$

$$Z = 27$$

$$Z_{eff} = Z - \sigma$$

$$= 27 - 23.10$$

$$Z_{eff} = 3.9$$

(9)

(4)

Hybridization in XeF_6 molecule

Repeat

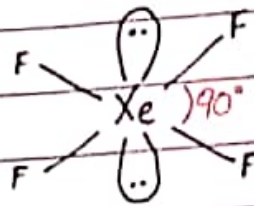
(5)

Structure according to VSEPR model

(i) AB_4E_2

$$\text{Bp} = 4, \text{Sp} = 2$$

e.g. $\text{XeF}_4, \text{ICl}_4, \text{BrF}_4$

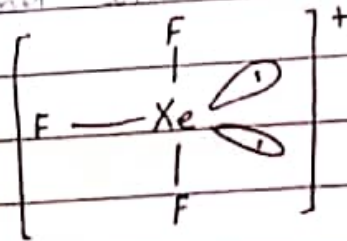


Square planar geometry

(ii) AB_3E_3

$$\text{Bp} = 3, \text{Sp} = 3$$

The trifluoroxenate(II) anion, XeF_3^- has been investigated as a possible first example of an AB_3E_3 molecule, having 6 $\bar{\text{e}}\text{pairs}$ in an octahedral arrangement with both 3 lp and 3 ligand in a T-shaped orientation



Short (2018)

(1)

(1)

Hybrid orbital exhibit better overlap than simple atomic orbitals.

Hybridization is a concept of mixing of two or more atomic orbitals of comparable energy, followed by their splitting into an equal no. of orbital with mixed character. The new orbitals are called hybrid orbitals. The energy of hybrid orbitals are always less than those of parent (atomic) orbitals. Hence they are more stable as compared to atomic orbital from which they are created. They undergo better overlap and form stronger bonds than the atomic orbitals.

(2)

Differentiate b/w

Polarizing power

Polarizability

— "When anion A^- is distorted by cation C^+ , this phenomenon is called distortion or polarization of A^- by C^+ "

If the degree of polarization is quite small, an ionic bond is formed, while if degree of polarization is large, a covalent bond results.

— "The ability of cation to distort

An anion is known as its polarizability⁽¹¹⁾ power and the tendency of the anion to become polarized by the cation is known as its Polarizability.



No polarization



Polarized anion

(3)

Fajan's rule:

Repeat

(4)

Calculate Z_{eff} for $4s-e$ in Cobalt ($Z = 27$)

Repeat

(5)

Can Lanthanum ion exist in +4 oxidation state? If yes then how, if no then why?

No, Lanthanum ion doesn't exist in +4 oxidation state.

Lanthanum has electronic configuration $[Xe] 4f^1 5s^2$ and forms La^{3+} ion by the removal of 3 electrons (two from $5s$ & one from $4f$) leaving behind stable noble gas configuration. To form La^{4+} ion stable noble gas configuration will have to be distributed which is impossible. Hence La^{4+} ion cannot exist.

(6)

(12)

What is 4c-3e bond? Give example

Generally there are two types of covalent bonds. One is the ordinary two centre, two electron (2c-2e) bond, and the second is the three centre four electron (3c-4e) bond, formed from a p-orbital of the central atom and the s-orbital of two other atoms.

For example:

